

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Use Laplace Transforms to Solve Non-Homogeneous ODEs

1. 1st-order Non-Homogeneous Linear ODEs

Non-homogeneous 1st-order ODEs can be solved using Laplace transforms.

Consider a non-homogeneous 1st-order ODE with an initial condition,

$$y' + py = f(t); \quad y(0) = y_0$$

where p is a constant and $f(t) \neq 0$. Then, we apply the notation $\mathcal{L}$ and linearity to write the ODE as

$$\mathcal{L}\{y'\} + p\mathcal{L}\{y\} = \mathcal{L}\{f(t)\}.$$

Then, we apply the table of transforms to transform the ODE from the t -domain to the s -domain.

- a. Consider this ODE with an initial condition: $y' + 2y = t; \quad y(0) = 1$.

The following tasks outlines the method solving the specific solution to the given non-homogeneous 1st-order ODE.

Task	Show work here
Let $\mathcal{L}\{y(t)\} = Y(s)$. Apply the Laplace transform $\mathcal{L}$ notation.	
Use the table of transforms to transform the ODE from the t -domain to the s -domain.	
Solve for $Y(s)$.	
Use the table again to apply the inverse Laplace transform $\mathcal{L}^{-1}$ to solve for $y(t)$. Make sure to use the correct notation.	

- b. Verify the specific solution determined in Part (a) by using the integrating factors method.

2. 2nd-order Non-Homogeneous Linear ODEs

Non-homogeneous 2nd-order ODEs can be solved using Laplace transform.

Consider a non-homogeneous 2nd-order ODE with an initial condition,

$$y'' + py' + qy = f(t); \quad y(0) = a_0, \quad y'(0) = a_1$$

where p and q are constants and $f(t) \neq 0$. Then, we apply the notation $\mathcal{L}$ and linearity to write the ODE as

$$\mathcal{L}\{y''\} + p\mathcal{L}\{y'\} + q\mathcal{L}\{y\} = \mathcal{L}\{f(t)\}.$$

Then, we apply the table of transforms to transform the ODE from the t -domain to the s -domain.

- a. Consider this ODE with initial conditions: $y'' + 2y' - 3y = t$; $y(0) = 1$, $y'(0) = 0$.

The following tasks outlines the method solving the specific solution to the given non-homogeneous 2nd-order ODE.

Task	Show work here
Let $\mathcal{L}\{y(t)\} = Y(s)$. Apply the Laplace transform $\mathcal{L}$ notation.	
Use the table of transforms to transform the ODE from the t -domain to the s -domain.	
Solve for $Y(s)$.	
Write $Y(s)$ into a sum of partial fractions	
Use the table again to apply the inverse Laplace transform $\mathcal{L}^{-1}$ to solve for $y(t)$. Make sure to use the correct notation.	

- b. The non-homogeneous ODE in Part (a) can also be solved using the method we discussed before, which works by splitting the problem into two parts. First, we find the homogeneous solution y_h . Then, we find the particular solution y_p . By the superposition principle, the complete solution is the sum $y = y_h + y_p$.

Using your results from Part (a), discuss which terms in your final expression correspond to the homogeneous solution y_h and which terms correspond to the particular solution y_p .